

31. Construct a C program to simulate the First in First Out paging technique of memory management.

AIM :

Construct a C program to simulate the First in First Out paging technique of memory management.

CODE:

```
#include <stdio.h>
```

```
#define MAX_FRAMES 3
```

```
void printFrames(int frames[], int n)
```

```
{  
    int i;  
  
    for (i = 0; i < n; i++)  
    {  
        if (frames[i] == -1)  
        {  
            printf(" - ");  
        }  
        else  
        {  
            printf(" %d ", frames[i]);  
        }  
    }  
  
    printf("\n");  
}
```

```
int main()
{
    int referenceString[] = {7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2};
    int n = sizeof(referenceString) / sizeof(referenceString[0]);

    int frames[MAX_FRAMES];
    int framePointer = 0;
    int i, j;

    /* Initialize frames */
    for (i = 0; i < MAX_FRAMES; i++)
    {
        frames[i] = -1;
    }

    /* Print reference string */
    printf("Reference String: ");

    for (i = 0; i < n; i++)
    {
        printf("%d ", referenceString[i]);
    }

    printf("\n\n");

    printf("Page Replacement Order:\n");
```

```

/* FIFO Page Replacement */
for (i = 0; i < n; i++)
{
    int page = referenceString[i];
    int pageFound = 0;

    /* Check whether page is already in memory */
    for (j = 0; j < MAX_FRAMES; j++)
    {
        if (frames[j] == page)
        {
            pageFound = 1;
            break;
        }
    }

    /* Page fault */
    if (!pageFound)
    {
        printf("Page %d -> ", page);

        frames[framePointer] = page;

        framePointer = (framePointer + 1) % MAX_FRAMES;

        printFrames(frames, MAX_FRAMES);
    }
}

```

```
}  
}
```

```
return 0;
```

OUTPUT:

```
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
```

```
Page Replacement Order:
```

```
Page 7 -> 7 - -
```

```
Page 0 -> 7 0 -
```

```
Page 1 -> 7 0 1
```

```
Page 2 -> 2 0 1
```

```
Page 3 -> 2 3 1
```

```
Page 0 -> 2 3 0
```

```
Page 4 -> 4 3 0
```

```
Page 2 -> 4 2 0
```

```
Page 3 -> 4 2 3
```

```
Page 0 -> 0 2 3
```

```
...Program finished with exit code 0
```

```
Press ENTER to exit console.
```